MA1522 Linear Algebra for Computing

Lesson 10: Basis and Dimension

Announcement: Online Quiz 2

- **Quiz 2 covers Chapter 2**, Sections 2.1–2.10: matrix algebra, matrix equations, inverses and elementary matrices, LU factorization, and determinants.

- It opens tomorrow, **Saturday 12 September, 0:00**, and is due **Sunday 13 September, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson10` (e.g. `A1234567Z_lesson10.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 10**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 3.6 Basis

Definition

Let V be a subspace of $\mathbb{R}^n$. A set $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \subseteq V$ is a **basis** for V if

(i) S spans V, that is $\text{span}(S) = V$, and

(ii) S is linearly independent.

Theorem

Suppose S is a basis for V. Then every vector $\mathbf{v} \in V$ can be written as a linear combination of vectors in S uniquely.

Section 3.7 Dimension

Definition

Let V be a subspace of $\mathbb{R}^n$. The **dimension** of V, written $\dim(V)$, is the number of vectors in any basis of V.

Theorem

Let A be an $m \times n$ matrix. The number of non-pivot columns in the reduced row-echelon form of A is the dimension of the solution space $V = \{ \mathbf{u} \in \mathbb{R}^n \mid A\mathbf{u} = \mathbf{0} \}$.

If $S \subseteq V$ and $|S| > \dim(V)$, then S is linearly dependent; if $|S| < \dim(V)$, then S cannot span V.

Question 1

Let V be the solution space of the homogeneous linear system

$$\begin{cases} x_1 + x_2 + 3x_3 + 2x_4 = 0 \\ x_1 - x_2 + x_3 - 4x_4 = 0 \\ 2x_1 + 3x_2 + 7x_3 + 7x_4 = 0 \end{cases}$$

Find a basis S for V, and find $\dim(V)$.

Question 1 — solution

Reduce the coefficient matrix.

$$\begin{pmatrix} 1 & 1 & 3 & 2 \\ 1 & -1 & 1 & -4 \\ 2 & 3 & 7 & 7 \end{pmatrix} \xrightarrow{R_2 - R_1,\ R_3 - 2R_1} \begin{pmatrix} 1 & 1 & 3 & 2 \\ 0 & -2 & -2 & -6 \\ 0 & 1 & 1 & 3 \end{pmatrix} \xrightarrow[R_1 - R_2]{-\frac{1}{2}R_2,\ R_3 - R_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Columns 3 and 4 are non-pivot, so $\dim(V) = 2$. Setting $x_3 = s$ and $x_4 = t$ gives $x_1 = -2s + t$ and $x_2 = -s - 3t$, so the general solution is

$$s \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{R}. \quad \checkmark$$

Therefore $S = \{\mathbf{s}_1, \mathbf{s}_2\}$ with $\mathbf{s}_1 = (-2, -1, 1, 0)^T$ and $\mathbf{s}_2 = (1, -3, 0, 1)^T$ is a basis for V, and $\dim(V) = 2$. □

Section 3.6 Basis for $\mathbb{R}^n$

Corollary

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a subset of $\mathbb{R}^n$ and let $A = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then S is a basis for $\mathbb{R}^n$ if and only if $k = n$ and A is invertible.

So for a square matrix A of order n, the list of equivalent statements from Chapter 2 gains two more entries:

(x) the columns of A are linearly independent;

(xi) the columns of A span $\mathbb{R}^n$.

Question 2

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$ be sets of vectors in $\mathbb{R}^4$, and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 \end{pmatrix}$. Suppose that

$$\operatorname{rref}(A) = \begin{pmatrix} 1 & 0 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 2 \end{pmatrix}.$$

Which of the following are true? Justify each answer.

(a) S is linearly independent.

(b) T is a basis for $\mathbb{R}^4$.

(c) $S \cup \{\mathbf{w}_1\}$ is a basis for $\mathbb{R}^4$.

(d) $S \cup \{\mathbf{w}_2\}$ is a basis for $\mathbb{R}^4$.

Question 2 — solution: (a) and (b)

Throughout: row reduction acts on rows, and $\mathrm{rref}(A) = EA$ for a single invertible E, so any selection of columns of $\mathrm{rref}(A)$ is row equivalent to the same selection of columns of A. One reduction answers a question about every sub-collection of the seven vectors at once.

(a) **True.** Columns $1, 2, 3$ of $\mathrm{rref}(A)$ are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, so $\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix}$ is row equivalent to a matrix with no non-pivot column. Hence S is linearly independent. (It is not a basis for $\mathbb{R}^4$: it has only three vectors.)

(b) **True.** Columns 4 to 7 give a matrix row equivalent to $\begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 \end{pmatrix}$, and its second column is $\mathbf{e}_4$, so

$$\det \begin{pmatrix} 2 & 0 & 1 & 3 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 2 \end{pmatrix} = \det \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix} = -9 \neq 0.$$

Row equivalent square matrices are invertible together, so $\begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 \end{pmatrix}$ is invertible and T is a basis for $\mathbb{R}^4$.

Question 2 — solution: (c) and (d)

(c) **False.** Column 4 is a non-pivot column, and it lists the coefficients

$$\mathbf{w}_1 = 2\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3. \quad \checkmark$$

So $\mathbf{w}_1 \in \operatorname{span}(S)$ and $S \cup \{\mathbf{w}_1\}$ is linearly dependent. It is not a basis.

(d) **True.** Column 5 is a pivot column, so columns $1, 2, 3, 5$ of $\operatorname{rref}(A)$ are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4$:

$$\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{w}_2 \end{pmatrix} \longrightarrow I_4,$$

so that matrix is invertible and $S \cup \{\mathbf{w}_2\}$ is a basis for $\mathbb{R}^4$.

The three vectors of S have been **extended to a basis** of $\mathbb{R}^4$ by one vector read off a single reduced row-echelon form. □

Question 3

Let V be the following subset of $\mathbb{R}^4$:

$$V = \left\{ \begin{pmatrix} a+d \\ a+b+2d+e \\ b+c+d+2e \\ c+e \end{pmatrix} \;\middle|\; a, b, c, d, e \in \mathbb{R} \right\}.$$

1. Show that V is a vector space, and find a basis for V and $\dim(V)$.

2. Let $\mathbf{s}_1 = (1, 1, 1, 1)^T$, $\mathbf{s}_2 = (1, 2, 2, 1)^T$ and $\mathbf{s}_3 = (0, 1, 0, -1)^T$. Show that $S = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$ is also a basis for V.

3. Find a set T of vectors in $\mathbb{R}^4$ such that $S \cup T$ is a basis for $\mathbb{R}^4$.

Question 3 — solution: part 1

Collect the coefficient of each parameter: $V = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_5\}$ with

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \ \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \ \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \ \mathbf{u}_4 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \ \mathbf{u}_5 = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}.$$

A linear span is a subspace of $\mathbb{R}^4$, so V is one. Now reduce.

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 - R_3]{R_2 - R_1, \ R_3 - R_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Columns $1, 2, 3$ are the pivot columns, so $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a basis for V and $\dim(V) = 3$. The other two columns say $\mathbf{u}_4 = \mathbf{u}_1 + \mathbf{u}_2$ and $\mathbf{u}_5 = \mathbf{u}_2 + \mathbf{u}_3$. ✓ □

Question 3 — solution: part 2

First, $S \subseteq V$: taking (a, b, c, d, e) to be $(1, 0, 1, 0, 0)$, $(1, 1, 1, 0, 0)$ and $(0, 1, -1, 0, 0)$ gives $\mathbf{s}_1$, $\mathbf{s}_2$ and $\mathbf{s}_3$. ✓ Hence $\text{span}(S) \subseteq V$.

Now stand $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3$ first and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ after them, and reduce once.

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & -1 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Columns $1, 2, 3$ are pivot columns, so S is linearly independent. Columns $4, 5, 6$ are not, and they give $\mathbf{u}_1 = 2\mathbf{s}_1 - \mathbf{s}_2 + \mathbf{s}_3$, $\mathbf{u}_2 = -\mathbf{s}_1 + \mathbf{s}_2$, $\mathbf{u}_3 = -\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3$ ✓, so $V \subseteq \text{span}(S)$.

Hence $\text{span}(S) = V$ and S is linearly independent: S is a basis for V. □

Question 3 — solution: part 3

$\dim(V) = 3$, so S is one vector short of a basis for $\mathbb{R}^4$. Adjoin the whole standard basis and reduce.

$$\left(\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|cccc} 1 & 0 & 0 & 0 & 2 & -3 & 2 \\ 0 & 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{array}\right).$$

The pivot columns are $1, 2, 3, 4$, so $\{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \mathbf{e}_1\}$ is linearly independent; four independent vectors in $\mathbb{R}^4$ form a basis. So $T = \{\mathbf{e}_1\} = \{(1, 0, 0, 0)^T\}$.

Check: $\det\begin{pmatrix} \mathbf{s}_1 & \mathbf{s}_2 & \mathbf{s}_3 & \mathbf{e}_1 \end{pmatrix} = -\det\begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 0 \\ 1 & 1 & -1 \end{pmatrix} = 1 \neq 0.$ ✓ □

Challenge

Let V be a subspace of $\mathbb{R}^n$, let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for V, and let $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\} \subseteq V$ be a set of k vectors in V.

(a) Show that if T is linearly independent, then $\text{span}(T) = \text{span}(S) = V$.

(b) Show that if $\text{span}(T) = V$, then T is linearly independent.

In particular, T is linearly independent if and only if T spans V: for a set of exactly k vectors in V, either one on its own already makes T a basis.

Next lesson

- **Sections 3.6–3.8 Coordinates, Dimensions, Transition Matrices**

Please watch the video clips for these sections beforehand.

Challenge — solution: the set-up and (a)

Every $\mathbf{w}_j$ lies in $V = \operatorname{span}(S)$, so there are scalars a_{ij} with

$$\mathbf{w}_j = a_{1j}\mathbf{u}_1 + a_{2j}\mathbf{u}_2 + \cdots + a_{kj}\mathbf{u}_k, \qquad j = 1, \ldots, k.$$

Let $A = (a_{ij})$ be the $k \times k$ matrix whose jth column holds the coefficients of $\mathbf{w}_j$. For any $\mathbf{c} = (c_1, \ldots, c_k)^T$, collecting the coefficient of each $\mathbf{u}_i$ gives

$$c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = (A\mathbf{c})_1\mathbf{u}_1 + \cdots + (A\mathbf{c})_k\mathbf{u}_k. \qquad (*)$$

(a) Let T be linearly independent. If $A\mathbf{c} = \mathbf{0}$ then $(*)$ gives $c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = \mathbf{0}$, so $\mathbf{c} = \mathbf{0}$. Thus $A\mathbf{x} = \mathbf{0}$ has only the trivial solution and A is invertible.

Now $\operatorname{span}(T) \subseteq V$ because each $\mathbf{w}_j \in V$. Conversely let $\mathbf{v} \in V$. As S spans V, $\mathbf{v} = b_1\mathbf{u}_1 + \cdots + b_k\mathbf{u}_k$; put $\mathbf{c} = A^{-1}\mathbf{b}$, and $(*)$ gives $c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = \mathbf{v}$. So $V \subseteq \operatorname{span}(T)$ and $\operatorname{span}(T) = \operatorname{span}(S) = V$. □

Challenge — solution: (b)

(b) Let $\operatorname{span}(T) = V$. Each $\mathbf{u}_i$ lies in $V = \operatorname{span}(T)$, so there are scalars b_{ji} with

$$\mathbf{u}_i = b_{1i}\mathbf{w}_1 + b_{2i}\mathbf{w}_2 + \cdots + b_{ki}\mathbf{w}_k, \qquad i = 1, \ldots, k;$$

let $B = (b_{ji})$. Applying $(*)$ to the ith column of B,

$$\mathbf{u}_i = (AB)_{1i}\mathbf{u}_1 + (AB)_{2i}\mathbf{u}_2 + \cdots + (AB)_{ki}\mathbf{u}_k.$$

But S is a basis, so this expression for $\mathbf{u}_i$ is unique — and $\mathbf{u}_i = 0\mathbf{u}_1 + \cdots + 1\mathbf{u}_i + \cdots + 0\mathbf{u}_k$. Hence the ith column of AB is $\mathbf{e}_i$ for every i, that is $AB = I_k$, and a square matrix with a right inverse is invertible. So A is invertible.

Finally, if $c_1\mathbf{w}_1 + \cdots + c_k\mathbf{w}_k = \mathbf{0}$ then $(*)$ gives $(A\mathbf{c})_1\mathbf{u}_1 + \cdots + (A\mathbf{c})_k\mathbf{u}_k = \mathbf{0}$; since S is linearly independent, $A\mathbf{c} = \mathbf{0}$; since A is invertible, $\mathbf{c} = \mathbf{0}$. So T is linearly independent. □

Both halves are the same three lines with the roles of S and T exchanged.